

MID TERM EXAMINATION

Subject: Linked Lists, Stacks, Queues

Paper: Mid Term Examination - Set A | Duration: 3 Hours | Max Marks: 37 | Date: 11-03-2026

Total Questions: 11 | Answer all questions.

Section 1: MCQ (6 Questions - 12 Marks)

Q1. [2 marks] (Medium)

Here are 10 unique exam questions for the Data Structures (CS101) Mid Term Examination:

Q2. [2 marks] (Easy)

In a singly linked list, which of the following operations takes $O(1)$ time if a pointer to the previous node is available?

Options:

A) Deleting the last node B) Inserting a node at the beginning C) Deleting the current node D) Searching for a specific value

Q3. [2 marks] (Medium)

Which of the following best describes the principle used for data access in a stack?

Options:

A) First-In, First-Out B) Last-In, First-Out C) Random Access D) Best-Fit

Q4. [2 marks] (Easy)

Which of the following is the primary characteristic of a Queue data structure?

Options:

A) Last-In, First-Out (LIFO) B) First-In, First-Out (FIFO) C) Random Access D) Hierarchical Storage

Q8. [2 marks] (Medium)

In a circular queue implemented using an array of size N , if $\text{front} = (\text{rear} + 1) \% N$, what does this state typically indicate?

Options:

A) The queue is empty. B) The queue is full. C) The queue has only one element. D) The queue is in an invalid state.

Q10. [2 marks] (Hard)

Consider a doubly linked list. If you have a pointer to a specific node X , what is the time complexity to insert a new node Y before node X ?

Options:

A) $O(1)$ B) $O(n)$ C) $O(\log n)$ D) $O(n \log n)$

Section 2: Short Answer (5 Questions - 25 Marks)

Q5. [5 marks] (Medium)

Describe the two main components of a node in a typical singly linked list. How do these components facilitate the list's structure?

Q6. [5 marks] (Hard)

Differentiate between a Stack Overflow and Stack Underflow condition. When do these typically occur?

Q7. [5 marks] (Easy)

List two advantages of using a linked list over a fixed-size array for storing a collection of elements that may frequently change in size.

Q9. [5 marks] (Medium)

Describe the main difference between a singly linked list and a doubly linked list in terms of node structure and traversal capabilities.

Q11. [5 marks] (Hard)

Why is a Circular Queue often preferred over a simple Linear Queue when both are implemented using arrays? Explain the drawback of a linear queue that a circular queue specifically addresses.

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